DATA SCIENCE
&
ARTIFICIAL INTELLIGENCE



Probability and Statistics
Statistics - 2

Lecture No. 08



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Recap of previous lecture



Topic

t - TEST

Topics to be Covered



Topic

"CHI-SQUARE TEST"

(large sample test is for $n \geq 50$)

CHI-SQUARE DISTRIBUTION

Let $x_1, x_2, x_3 \ldots\ldots x_n$ are (Indel) Normal Random Variables with mean μ & Var σ^2

then $z_i = \frac{x_i - \mu}{\sigma}$ is $z_1, z_2, z_3, \ldots z_n$ are standard Normal Variables with mean 0 & Variance 1

and $z_1^2, z_2^2, z_3^2, \ldots\ldots, z_n^2$ are called Chi-square Variables.

and the Distribution $z_1^2 + z_2^2 + z_3^2 + \ldots\ldots + z_n^2$ is called Chi Square Distribution.

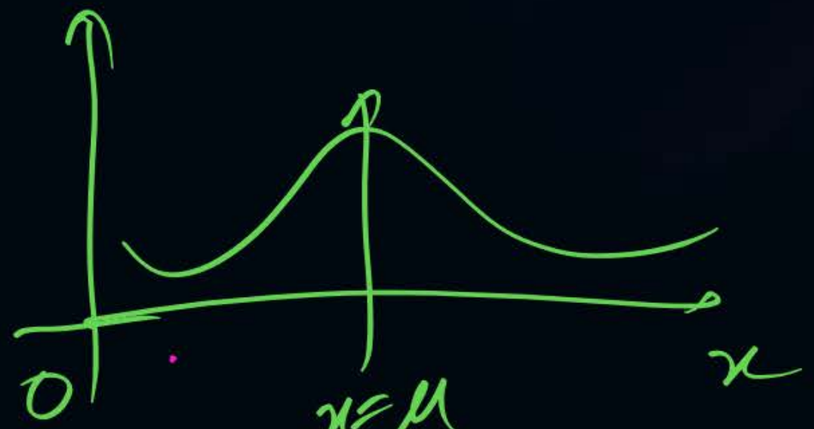
in Short, $z_i^2 = \left(\frac{x_i - \mu}{\sigma}\right)^2$ are called Chi Sq. Variables.

$$\chi^2 = \sum_{i=1}^{n} z_i^2 = \sum_{i=1}^{n} \left(\frac{x_i - \mu}{\sigma}\right)^2$$

& it has only one parameter = n

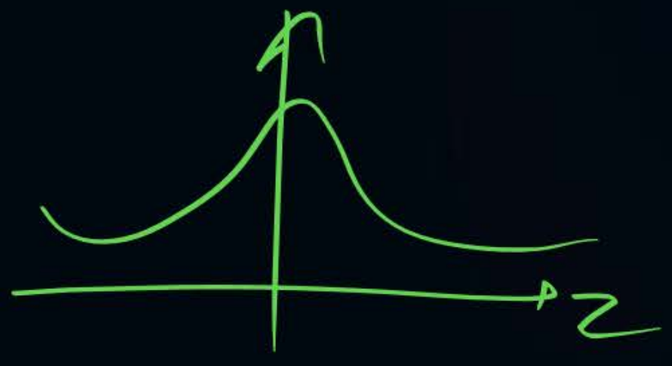
& $Df(D) = n - 1$

$0 \leq \chi^2 < \infty$



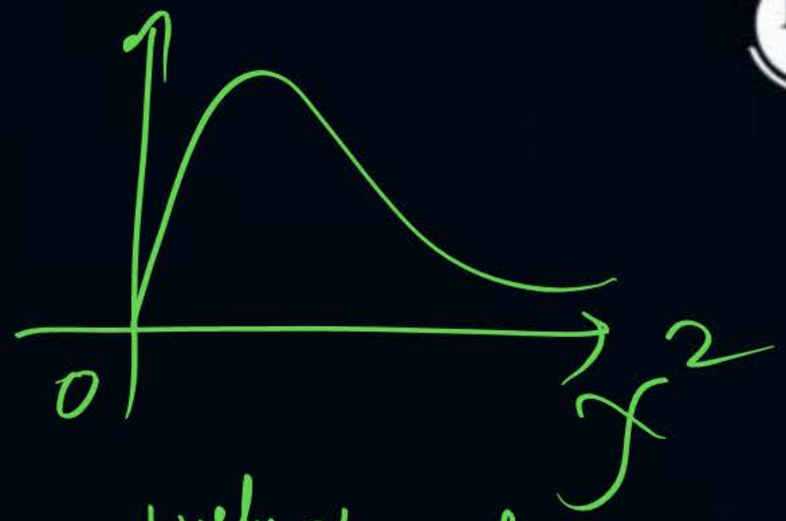
$x = \mu$

Symmetrical Distribution



$Z = 0$

Symmetrical dist.



+vely skewed.

Note ① Sample Values ≃ Experimental Values ≃ Observed Values ($= O_i$) ≃ Exact Values

② Population Values ≃ Theoretical Values ≃ Expected Values ($= E_i$) ≃ Approx Values

with the help of Chi sq. test, we can solve following types of Questions:

Type (1) Significance of Population Variance →

$H_0 : \sigma^2 = \sigma_0^2$, $H_1 : \sigma^2 \neq \sigma_0^2$, & Sample Variance $s^2 = \frac{\sum (x-\bar{x})^2}{n}$

$$\chi^2 = \sum \left(\frac{x-\bar{x}}{\sigma}\right)^2 = \frac{\sum (x-\bar{x})^2}{\sigma^2} = \frac{ns^2}{\sigma^2}$$ Ham.

Type II: Significance of Goodness of fit →

H_0 : there is (No) Significant Difference b/n Observed Values (O_i) & Expected Values (E_i)

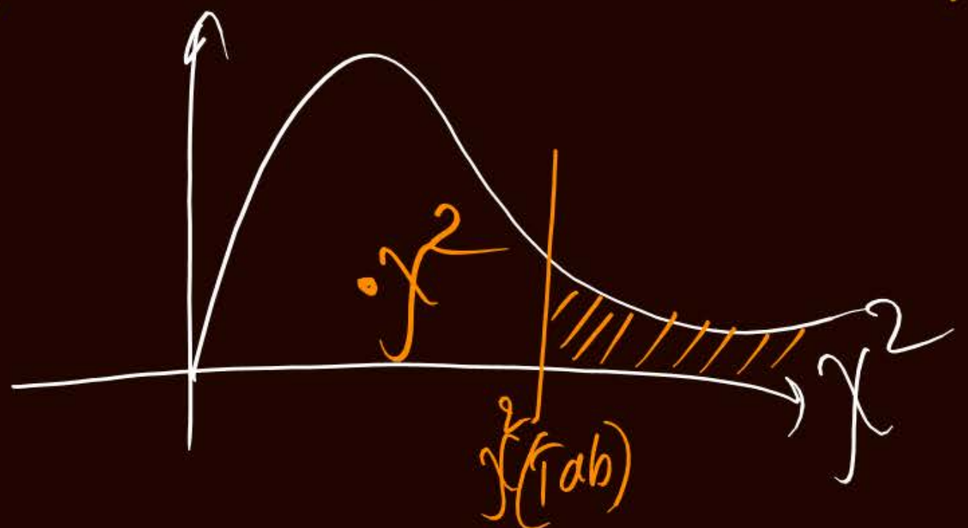
= sample values = population values

H_1 : there is a " " " " " " " "

Here we have one constraint ; $\sum O_i = \sum E_i$ & $Df = n - 1$

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

if $\chi^2(Cal) \leq \chi^2(Tab)$ & Accept H_0



Type 3 → Significance of Independence of Attributes →

H_0 : "there is No Relationship b/n two Attributes" i.e Two Attributes are Ind

H_1 : there is a relationship " " " " i.e " " " not Ind

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad \text{where } E_i = \frac{R_i \times C_j}{Total} = \frac{\text{Row Total} \times \text{Column Total}}{Total}$$

Contigentency Table = $R \times C \Rightarrow Df = (R-1)(C-1)$

Degree of Freedom → DF = Total No. of observations − No. of constraints imposed

Using χ^2 distribution, D.f for various distribution will be

(1) Binomial Dist → DF = $(n-1)$ $(\because \sum O_i = \sum E_i)$

(2) Poisson Dist → DF = $(n-2)$ $(\because \sum O_i = \sum E_i$ & Mean = $\lambda)$

(3) Normal Dist → DF = $(n-3)$ $(\because \sum O_i = \sum E_i$ & Mean = μ, SD = $\sigma)$

#Q. A random sample of size 25 has sample standard derivation = 9. Test the Hypothesis that population standard derivation is 10.5.

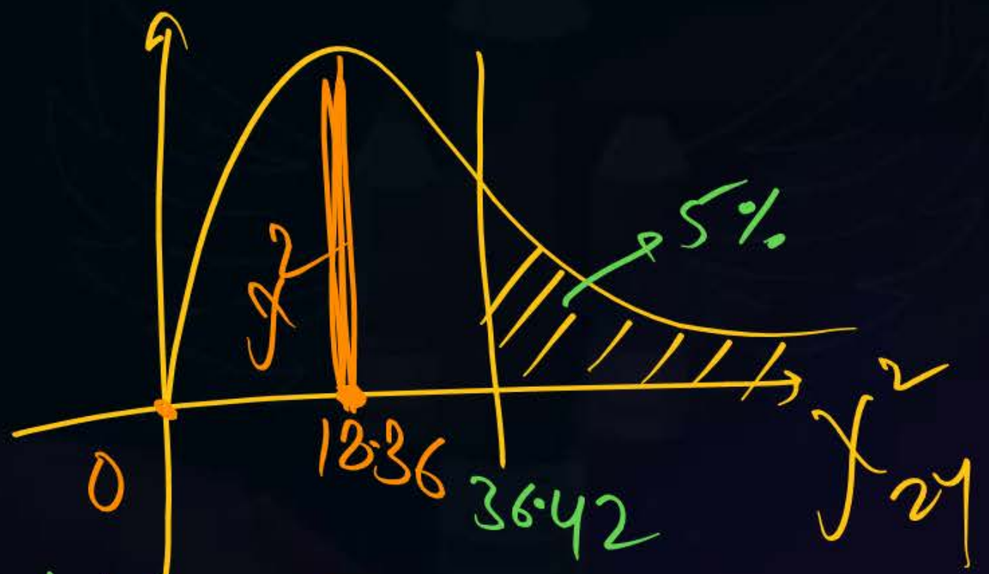
Type 1

Given $x^2_{24}(0.05) = 36.42$

$n = 25,\ s = 9,\ \sigma = ?$. $H_0 : \sigma = 10.5$, $H_1 : \sigma \neq 10.5$

$\nu = (n-1) = 24$

$$\chi^2 = \frac{ns^2}{\sigma^2} = \frac{25 \times 81}{(10.5)^2} = 18.36$$



$\because \chi^2_{(Calculated)} < \chi^2_{(Tabulated)}$ so Accept H_0 & Reject H_1

#Q. (i) A random sample of size 20 has variance 25. If the population standard derivation is 8 calculated x^2

(ii) Also calculated the hypothesis that $\sigma = 8$ for 95% confidence region.

Given $x^2_{19}(0.05) = 30.14$

$n=20,\ s^2=25,\ \sigma=8,\ \chi^2 = \frac{ns^2}{\sigma^2} = \frac{20\times 25}{(8)^2} = \frac{500}{64} = 7.81$

$\nu = n-1 = 19$

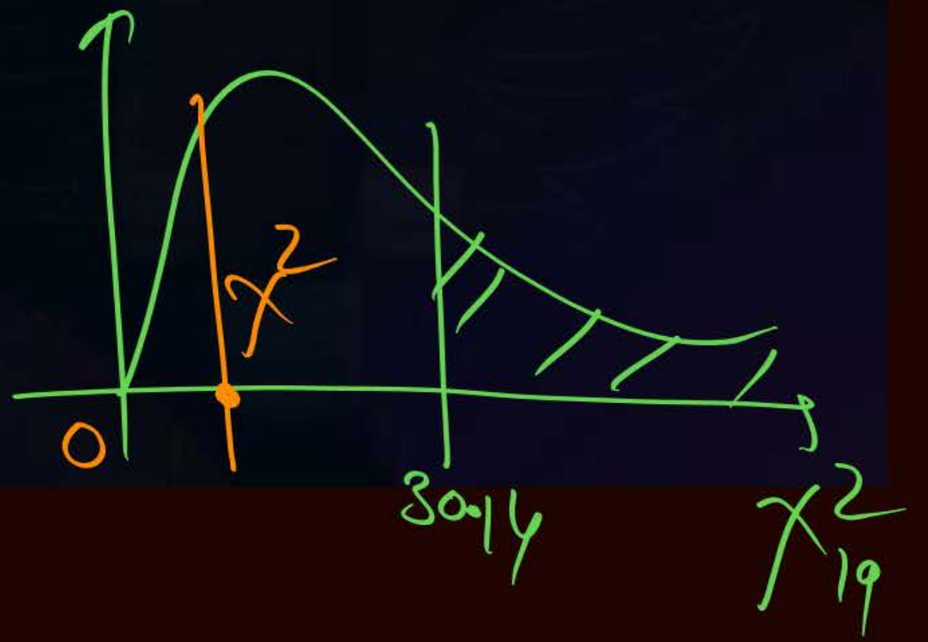
(ii) $H_0 : \sigma = 8$

$H_1 : \sigma \neq 8$

$\therefore \chi^2_{cal} < \chi^2_{Tab}$

$\therefore$ Accept H_0

& Reject H_1



#Q. The following table shows the numbers of car accidents that occurred during the week. Test whether the accident are uniformly distributed over the week.

Type 2

Given $x_6^2(0.05) = 12.592$

Day	Mon	Tue	Wed	Thurs	Fri	Sat	Sun	Total accident
Number of accident	14	18	12	11	15	14	14	98

$O_i =$

Sol: If Accidents are uniformly distributed then

Expected No of Accidents per day $E_i = \frac{\text{Total Accidents}}{7} = \frac{98}{7} = 14$

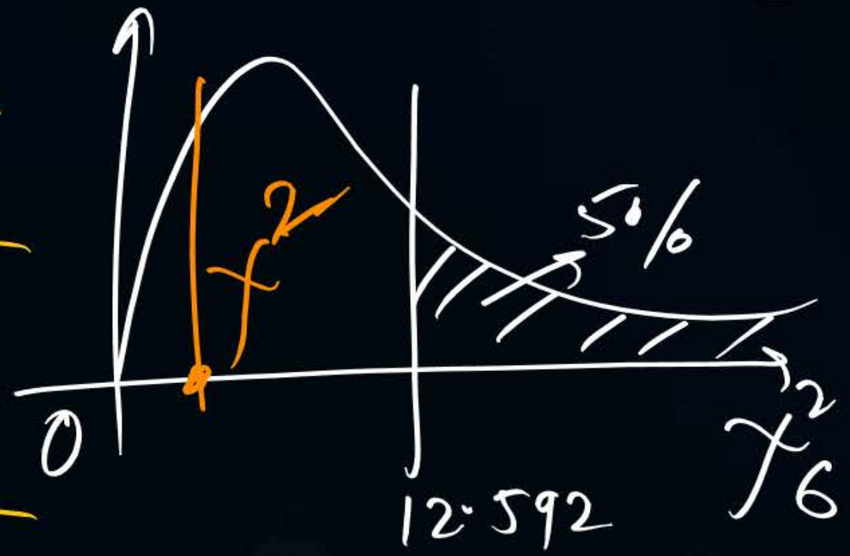
H_0: there is no difference b/n O_i & E_i i.e Accidents are Uniformly distributed.

H_1: there is a difference b/n O_i & E_i i.e " are not " "

	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
M	14	(14)	0	0	0
T	18	(14)	4	16	16/14
W	12	14	−2	4	4/14
Th	11	14	−3	9	9/14
f	15	14	1	1	1/14
Sat	14	14	0	0	0
S	14	14	0	0	0
	$\sum O_i = 98$	$\sum E_i = 98$			

$\because n = 7 \Rightarrow \nu = 6$

$$\sum_{i=1}^{7} \frac{(O_i - E_i)^2}{E_i} = 0 + \frac{16+4+9+1+0+0}{14} = \frac{30}{14} = 2.14$$



So Accept H_0.
& Reject H_1
ie Accidents are U-Dist.

#Q. A die is thrown 276 times and frequencies for various outcomes are

f(1) = 40, f(2) = 32, f(3) = 29, f(4) = 59, f(5) = 57, f(6) = 59. Test whether die is Biased or not?

it is given that $x_5^2(0.05) = 11.09$

Sol: $O_1 = 40, O_2 = 32, O_3 = 29, O_4 = 59, O_5 = 57, O_6 = 59$ & $\sum O_i = 276$

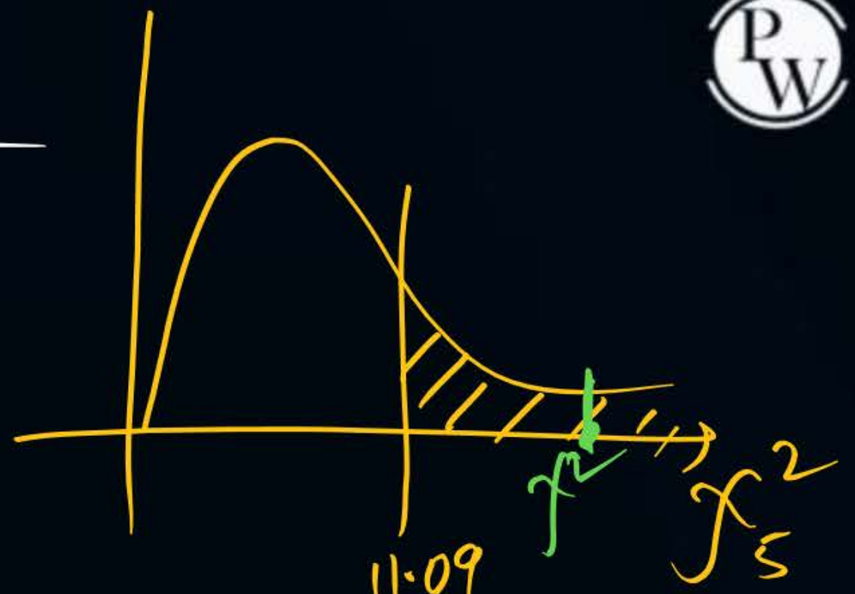
If Die is Unbiased then Expected frequency of each outcome $= \frac{276}{6} = 46$

$\rightarrow E_1 = 46, E_2 = 46, E_3 = 46, E_4 = 46, E_5 = 46, E_6 = 46$ & $\sum E_i = 276$

H_0 : there is No Significant Difference b/n O_i & E_i i.e Die is Unbiased

H_1 : there is Significant diff b/n O_i & E_i i.e Die is Biased

Outcome	O_i	E_i	$(O_i-E_i)^2$	$(O_i-E_i)^2/E_i$
1	46	46	36	36/46
2	32	46	196	196/46
3	29	46	289	289/46
4	59	46	169	169/46
5	57	46	121	121/46
6	59	46	169	169/46
	$\Sigma=276$	$\Sigma=276$		



$\therefore$ Reject H_0 & Accept H_1
i.e Die is BIASED.

$$\sum\frac{(O_i-E_i)^2}{E_i} = \frac{36+196+289+169+121+169}{46}$$

$$\chi^2 = 21.30$$

#Q. A car manufacture company produces four different color (Cars) in which 882 are white, 313 Grey, 287 Red and 118 Black. While at the beginning they had decided theoretically that they will produce white, ~~Green~~ (Grey), red, Black cars in the ratio 9 : 3 : 3 : 1 respectively. If $x_3^2(0.05) = 7.815$, ~~Enemies~~ (Establish) the correspondence between (Theory) and experiment. does the experimental result support the theory?

HW8

$n = 4, v = 3$

Colour	Oi	Ei
White	882	—
Grey	313	—
Red	287	—
Black	118	—

↓

YES

#Q. What are the expected frequency of 2 × 2 contingency table given below:

(1)

a	b	a+b
c	d	c+d
a+c	b+d	a+b+c+d

(2)

2	10	12
6	6	12
8	16	

$$\text{Expected Value} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Total}}$$

$$E(a) = \frac{(a+b)(a+c)}{(a+b+c+d)}$$

$$E(b) = \frac{(a+b)(b+d)}{(a+b+c+d)}$$

$$E(c) = \frac{(c+d)(a+c)}{(a+b+c+d)}$$

$$E(d) = \frac{(c+d)(b+d)}{(a+b+c+d)}$$

$$E(2) = \frac{12 \times 8}{24}$$

$$E(10) = \frac{12 \times 16}{24}$$

$$E(6) = \frac{12 \times 8}{24}$$

$$E(6) = \frac{12 \times 16}{24}$$

#Q. In an engineering college of Delhi, the IQ and economic condition at home of 1000 students are given in the table;

I.Q / Eco-condition	High	Low	Total
Rich	100	300	400
Poor	350	250	600
Total	450	550	1000

It is 2 × 2 contingency table

can we conclude any association between economic condition at home and IQ of the students. Given $\chi_1^2(0.05) = 3.84$

$E(250) = \frac{600 \times 550}{1000} = 330$

$E(100) = \frac{400 \times 450}{1000} = 180$, $E(300) = \frac{400 \times 550}{1000} = 220$, $E(350) = \frac{600 \times 450}{1000} = 270$

O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
100	180	$(80)^2$	$(80)^2/180 =$
300	220	$(80)^2$	$(80)^2/220 =$
350	270	$(80)^2$	$(80)^2/270 =$
250	330	$(80)^2$	$(80)^2/330 =$



$$\sum \frac{(O_i - E_i)^2}{E_i} = 107.7$$

H_0: there is No Relationship b/w IQ & Eco. Cond^n at Home i.e. Both are Ind

H_1: there is a Relationship b/w " " " " i.e. " " Dependent.

$\therefore (\text{Table})_{2\times 2}$ at 5% $= (2-1)(2-1) = 1 \times 1 = 1$



$\therefore \chi_1^2 (\text{Calculated}) > \chi^2 (\text{Tabulated})$

So Reject H_0

& Accept H_1

ie there is an association b/w IQ & Eco-Condition. at Home

#Q. From the following data, Find out whether there is any relationship between sex and preference of color;

Sex / Color	Males	Female	Total
Red	10	40	50
White	70	30	100
Green	30	20	50
Total	110	90	200

It is 3 × 2 contingency table so difference = (3 – 1) (2 – 1) = 2 and

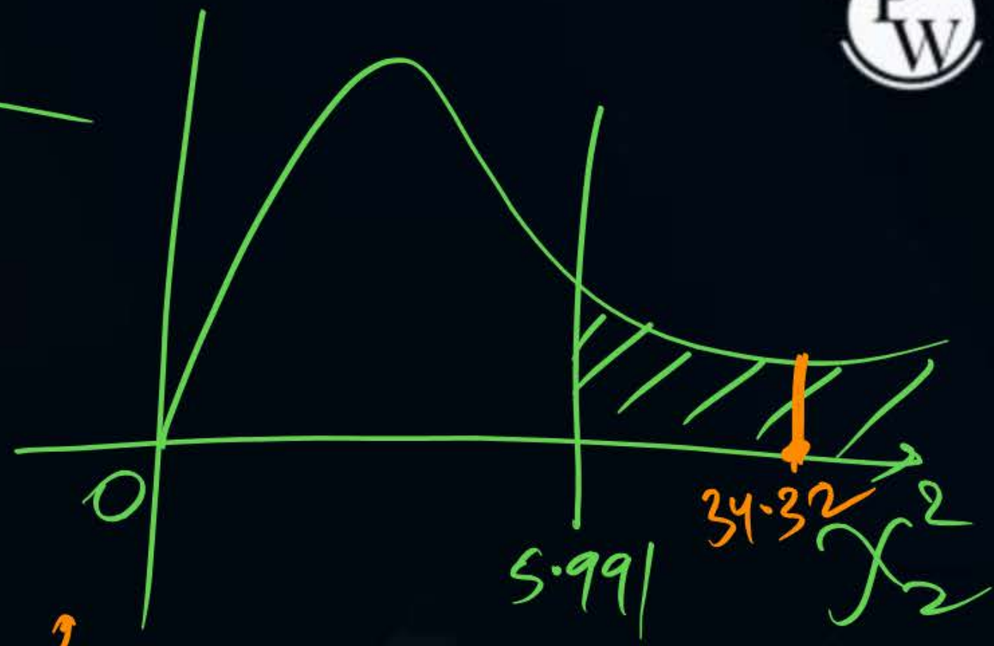
$x_2^2(0.05) = 5.991$

Df

$E(10) = \frac{50 \times 110}{200} = 27.5$, $E(40) = \frac{50 \times 90}{200} = 22.5$,

$E(70) = \frac{140 \times 110}{200} = 55$, $E(30) = \frac{140 \times 90}{200} = 45$, $E(30) = \frac{50 \times 110}{200} = 27.5$, $E(20) = \frac{50 \times 90}{200} = 22.5$

O_i	E_i	$(O_i-E_i)^2$	$(O_i-E_i)^2/E_i$
10	27.5	$(17.5)^2$	
40	22.5	$(17.5)^2$	
70	55	225	
30	45	225	
30	27.5	6.25	
20	22.5	6.25	



$\therefore \chi^2(\text{calculated}) > \chi^2(\text{Tabulated})$ & Reject H_0 & Accept H_1

$$\chi^2 = \sum \frac{(O_i-E_i)^2}{E_i} = 34.32$$

H_0: there is no Relationship b/w Gender & Preference of Colour i.e Both are Ind.
H_1: " is a " " " " " " Both are dependent.

#Q. In a survey of 200 boys of which 75 were intelligent, and remaining were unintelligent out of intelligent boys 40 had educated father while out of unintelligent boys 85 had uneducated fathers.

HWQ

Can we conclude that educated fathers have intelligent boys given that $x_1^2(0.05) = 3.841$

Sol: H_0 : there is no Relationship b/n Education of father & Intelligence of Boys.

H_1 : there is a Relationship " " " " "

Ans: Reject H_0 & Accept H_1

#Q. The number of screw declared fit or unfit by three inspectors x, y and z is shown in the following table.

HW

Inspectors	X	Y	z	Total
Fit screws	50	47	56	153
Unfit screw	5	14	8	27
Total	55	61	64	~~80~~ 180

Given $x_1^2(0.05) = 3.81$ and $2(0.05) = 5.99$ test the hypothesis that proportion of screws declare unfit by three inspectors are same.

Ans: Accept H_0 & Reject H_1

$r = \pm\sqrt{b_{yx}\, b_{xy}}$: $-1 \le r \le 1$

$\boxed{r^2 = b_{yx} \cdot b_{xy}}$: $0 \le r^2 \le 1$

THANK - YOU

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